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PYTHON / 240901502_data_aggrearation_group.ipynb

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300 lines (300 loc) · 8.7 KB

Preview Code Blame Raw

In [14]:
```
import pandas as pd
import numpy as np
df = pd.DataFrame([[1, 2, 3], [4, 5, 6], [7, 8, 9], [np.nan, np.nan, np.nan]], columns=['A', 'B', 'C'])
print("Original DataFrame\n",df)
print("\nAggregate function: aggregate()\n", df.aggregate(['sum']))
print("\nAggregate function: alias agg()\n", df.agg(['sum']))
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN

Aggregate function: aggregate()
         A     B     C
sum  12.0  15.0  18.0

Aggregate function: alias agg()
         A     B     C
sum  12.0  15.0  18.0
```

In [16]:
```
#Aggregate function to produce sum,min and max
a=df.agg(['sum', 'min', 'max'])
print("\nOriginal DataFrame\n",df)
print("\nAggregate function to produce sum, min and max\n",a)
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN

Aggregate function to produce sum, min and max
         A     B     C
sum  12.0  15.0  18.0
min   1.0   2.0   3.0
max   7.0   8.0   9.0
```

In [18]:
```
#Different aggregation per column
b=df.agg({'A': ['sum', 'min'], 'B': ['min', 'max']})
print("\nOriginal DataFrame\n",df)
print("\nDifferent aggregations per column\n",b)
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN

Different aggregations per column
         A    B
sum  12.0  NaN
min   1.0  2.0
max   NaN  8.0
```

In [20]:
```
#Aggregate different functions over the columns and rename the index of the resulting DataFrame
c=df.agg(x=('A', 'max'), y=('B', 'min'), z=('C', np.mean))
print("\nOriginal DataFrame\n",df)
print("\nAggregate different functions over the columns and rename the index of the resulting\nDataFrame\n",c)
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN

Aggregate different functions over the columns and rename the index of the resulting
DataFrame
      A    B    C
x  7.0  NaN  NaN
y  NaN  2.0  NaN
z  NaN  NaN  6.0
/tmp/ipykernel_586/326865355.py:2: FutureWarning: The provided callable <function mean at 0x7f4bb6331620> is currentl
y using Series.mean. In a future version of pandas, the provided callable will be used directly. To keep current beha
vior pass the string "mean" instead.
  c=df.agg(x=('A', 'max'), y=('B', 'min'), z=('C', np.mean))
```

In [22]:
```
d=df.agg("sum", axis="columns")
print("\nOriginal DataFrame\n",df)
print("\nAggregate-Sum: row wise\n",d)
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN
```

```
Aggregate-Sum: row wise
 0     6.0
1    15.0
2    24.0
3     0.0
dtype: float64
```

In [24]:

```
#Aggregate over the columns
e=df.agg("sum", axis="rows")
print("\nOriginal DataFrame\n",df)
print("\nAggregate-Sum: Column wise\n", e)
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN

Aggregate-Sum: Column wise
 A    12.0
B    15.0
C    18.0
dtype: float64
```

In [26]:

```
#Summary of aggregate function using describe
print("\nOriginal DataFrame\n",df)
print("\nSummary of aggregate functions using describe()\n",df.describe())
```

```
Original DataFrame
      A    B    C
0  1.0  2.0  3.0
1  4.0  5.0  6.0
2  7.0  8.0  9.0
3  NaN  NaN  NaN

Summary of aggregate functions using describe()
          A    B    C
count  3.0  3.0  3.0
mean   4.0  5.0  6.0
std    3.0  3.0  3.0
min    1.0  2.0  3.0
25%    2.5  3.5  4.5
50%    4.0  5.0  6.0
75%    5.5  6.5  7.5
max    7.0  8.0  9.0
```